TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

BMDS2013 Data Engineering

CLO2 ASSIGNMENT 202601

Team Reference : **G6-3**

Programme / Group : **RDS2 S3 / G6**

Declaration

- I confirm that I have read and complied with all the terms and conditions of Tunku Abdul Rahman University of Management and Technology's plagiarism policy.
- I declare that this assignment is free from all forms of plagiarism and for all intents and purposes is my own properly derived work.

Team members (***in alphabetical order of name*** as appears in your IC/Passport):

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Photo:	Photo:	Photo:	Photo:	Photo:
				

Room Occupancy and Usage Behaviour Analytics

1. Problem Description and Objectives

1.1. Problem Statement

University shared spaces require real-time occupancy insights to prevent overcrowding and improve space management. Building accurate prediction models for these areas involves integrating diverse IoT data, such as Wi-Fi connections, environmental sensors, and HVAC states.

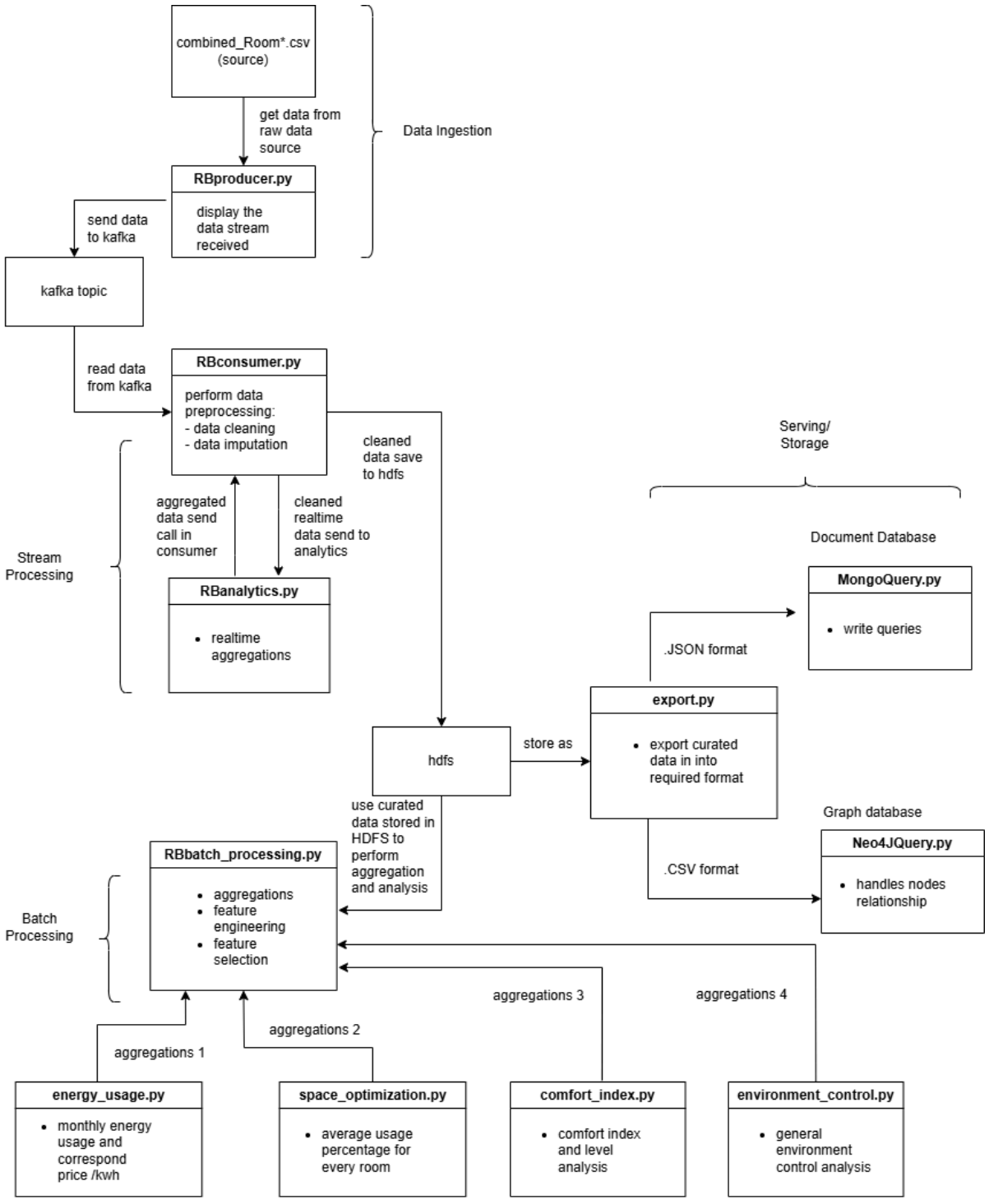
1.2. Target users

1.2.1. Students
1.2.2. Academic Staff
1.2.3. Department of Facility Management (DFM)

1.3. Key objectives

1.3.1. Design an automated, end-to-end architecture handling both real-time streaming and historical batch processing.smd
1.3.2. Ingest, clean, and synchronise diverse time-series data (Wi-Fi counts, HVAC states, environmental metrics) into appropiate format.
1.3.3. Store raw historical data in a distributed file system (HDFS)and processed, real-time room states in a NoSQL database for fast querying or future ML prediction.

2. Overall Architecture Diagram



3. Data Source(s) and Schemas

3.1. Data Source(s)

Data source type (API / crawled / public dataset):

- Public Dataset from Github (https://github.com/ideas-lab-nus/robod)

3.2. Input Schema

Column Name	Data typex	Description
timestamp	datetime	Time of measurement (5-minute interval)
voc	float	Volatile Organic Compounds level
sound_pressure_level	float	Noise Level
indoor_relative_humidity	float	Indoor humidity (%)
air_temperature	float	Indoor temperature (°C)
illuminance	float	Light intensity
pm2.5	float	Fine particulate matter
indoor_co2	float	Indoor Carbon dioxide level
wifi_connected_devices	integer	Number of connected devices
ceiling_fan_energy	float	Fan energy consumption
lighting_energy	float	Lighting energy usage
plug_load_energy	float	Plug load energy
chilled_water_energy	float	Water cooling energy
fcu_fan_energy	float	HVAC fan energy
temp_setpoint	float	Temperature setting
fcu_fan_speed	float	Fan speed
supply_air_pressure	float	HVAC pressure
supply_air_temperature	float	HVAC air temperature
barometric_pressure	float	Atmospheric pressure
dry_bulb_temp	float	Outdoor temperature
global_horizon_solar_radiation	float	solar_radiation
wind_direction	float	Wind direction
wind_speed	float	Wind Speed
outdoor_co2	float	Outdoor CO2
rainfall_raw	float	Rainfall
outdoor_relative_humidity	float	Outdoor humidity
occupant_presence	integer	Occupancy (0/1)
occupant_count	integer	Number of occupants

4. Stream Processing Logic

4.1. Kafka Topic(s) Consumed

- library_occupancy

4.2. Spark Streaming Input Format

- library_occupancy

```
root
 |-- key: binary (nullable = true)
 |-- value: binary (nullable = true)
 |-- topic: string (nullable = true)
 |-- partition: integer (nullable = true)
 |-- offset: long (nullable = true)
 |-- timestamp: timestamp (nullable = true)
 |-- timestampType: integer (nullable = true)
```

4.3. Streaming Transformations Applied

- Parsing JSON string payload into structured data columns (from_json)
- Casting string columns to specific numeric data types (float, int)
- Converting string timestamps to standard TimestampType (to_timestamp using "yyyy-MM-dd HH:mm XXX")
- Handling missing/null data by substituting default values (coalesce to 0.0 or 0)
- Rounding numeric metrics to two decimal places (round)
- Renaming columns to standardized metrics/units (alias to voc_ppb, temp_c, fan_e_kwh, etc.)
- Appending static category columns via lit() ("Indoor Environmental", "Energy", "HVAC Operations", "Occupancy/WiFi")
- Partitioning and sorting data using window functions (Window.partitionBy, orderBy)
- Applying custom analytical aggregations (Occupancy estimation, Energy usage analysis, IEQ monitoring

4.4. Output Sink(s)

- Console Sink:
    - Append mode (Categorised data streams with 10-second triggers)
    - Complete mode (Occupancy, Energy, and IEQ Analytics)

- HDFS Sink: Parquet format (Append mode, path: /user/hduser/occupancy/robod_curated_v2/ with checkpoints)
- MongoDB Atlas Sink: Exported Collections (cleaned_overall_sensor, moving_average_1hr, Room_1_sensor to Room_5_sensor) into the library database.

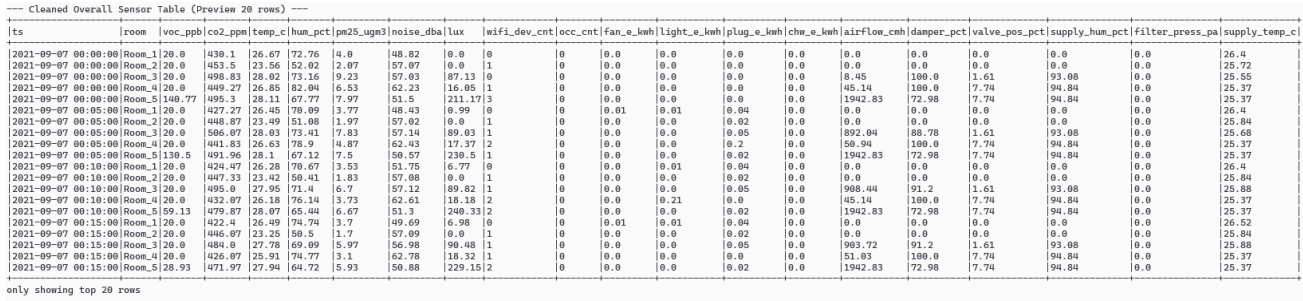
5. Batch Processing Logic

5.1. HDFS Input Path(s)

- /user/hduser/occupancy/robod_curated_v3/

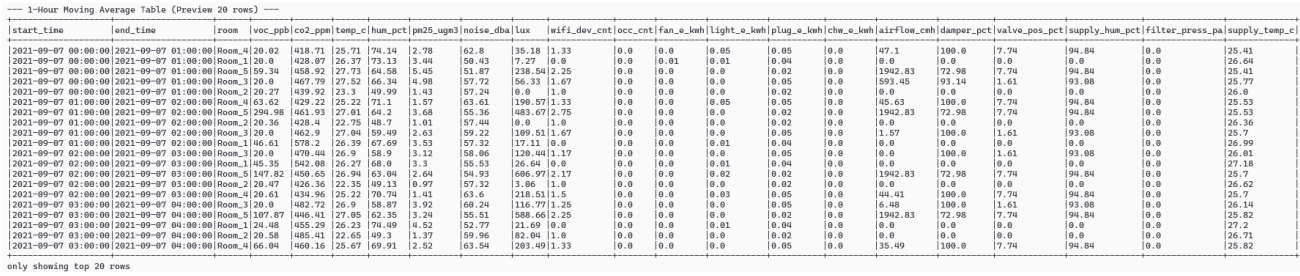
5.2. Batch Transformations or Aggregations

- Identifying and dropping numeric columns that contain entirely zero values
- Grouping data by 1-hour tumbling windows (window(col("ts"), "1 hour")) and room
- Calculating 1-hour moving averages for all active numeric sensor metrics (avg, round)
- Splitting window struct arrays into explicit start_time and end_time columns
- Filtering the overall moving average dataset into distinct subsets based on specific room IDs
- Extracting hour_of_day from timestamps to calculate overall average hourly occupancy
- Extracting day_of_week from timestamps to calculate overall average daily occupancy
- Applying conditional bucketing (when/otherwise) to categorize occupancy levels into "Low", "Medium", and "High"
- Applying custom sorting to enforce chronological day-of-week ordering (Monday to Sunday)
- Executing modular aggregations for Space Optimization, Environmental Control Summaries, Environmental Comfort Indexes, and Average Monthly Energy Consumption
- Before Spark DataFrame columns



```
--- Cleaned Overall Sensor Table (Preview 20 rows) ---
+-------------------+------+--------+--------+------+--------+---------+----------+-----+-------------+-------+-----------+-------------+-----------+-----------+-------------+-----------+-------------+-----------+-----------+-----------+
|ts                 |room  |voc_ppb |co2_ppm |temp_c|hum_pct |pm25_ugm3|noise_dba |lux  |wifi_dev_cnt |occ_cnt|fan_e_kwh  |light_e_kwh  |plug_e_kwh |chw_e_kwh  |airflow_cmh  |damper_pct |valve_pos_pct|supply_hum_pct|filter_press_pa|supply_temp_c|
+-------------------+------+--------+--------+------+--------+---------+----------+-----+-------------+-------+-----------+-------------+-----------+-----------+-------------+-----------+-------------+-----------+-----------+-----------+
|2021-09-07 00:00:00|Room_1|20.0    |430.1   |26.67 |72.76   |4.0      |48.82     |0.0  |0            |0      |0.0        |0.0          |0.0        |0.0        |0.0          |0.0        |0.0          |0.0           |0.0            |26.4         |
|2021-09-07 00:00:00|Room_2|20.0    |453.5   |23.56 |52.02   |2.07     |57.07     |0.0  |1            |0      |0.0        |0.0          |0.0        |0.0        |0.0          |0.0        |0.0          |0.0           |0.0            |25.72        |
|2021-09-07 00:00:00|Room_3|20.0    |498.83  |28.02 |73.16   |9.23     |57.03     |87.13|0            |0      |0.0        |0.0          |0.0        |0.0        |8.45         |100.0      |1.61         |93.08         |0.0            |25.55        |
|2021-09-07 00:00:00|Room_4|20.0    |449.27  |26.85 |82.04   |6.53     |62.23     |16.05|1            |0      |0.0        |0.0          |0.0        |0.0        |45.14        |100.0      |7.74         |94.84         |0.0            |25.37        |
|2021-09-07 00:00:00|Room_5|140.77  |495.3   |28.11 |67.77   |7.97     |51.5      |211.17|3           |0      |0.0        |0.0          |0.0        |0.0        |1942.83      |72.98      |7.74         |94.84         |0.0            |25.37        |
|2021-09-07 00:05:00|Room_1|20.0    |427.27  |26.45 |70.09   |3.77     |48.43     |0.99 |0            |0      |0.01       |0.01         |0.04       |0.0        |0.0          |0.0        |0.0          |0.0           |0.0            |26.4         |
|2021-09-07 00:05:00|Room_2|20.0    |448.87  |23.49 |51.08   |1.97     |57.02     |0.0  |1            |0      |0.0        |0.0          |0.02       |0.0        |0.0          |0.0        |0.0          |0.0           |0.0            |25.84        |
|2021-09-07 00:05:00|Room_3|20.0    |506.07  |28.03 |73.41   |7.83     |57.14     |89.03|1            |0      |0.0        |0.0          |0.05       |0.0        |892.04       |88.78      |1.61         |93.08         |0.0            |25.68        |
|2021-09-07 00:05:00|Room_4|20.0    |441.83  |26.63 |78.9    |4.87     |62.43     |17.37|2            |0      |0.0        |0.0          |0.2        |0.0        |50.94        |100.0      |7.74         |94.84         |0.0            |25.37        |
|2021-09-07 00:05:00|Room_5|130.5   |491.96  |28.1  |67.12   |7.5      |50.57     |230.5|1            |0      |0.0        |0.0          |0.02       |0.0        |1942.83      |72.98      |7.74         |94.84         |0.0            |25.37        |
|2021-09-07 00:10:00|Room_1|20.0    |424.47  |26.28 |70.67   |3.53     |51.75     |6.77 |0            |0      |0.0        |0.01         |0.04       |0.0        |0.0          |0.0        |0.0          |0.0           |0.0            |26.4         |
|2021-09-07 00:10:00|Room_2|20.0    |447.33  |23.42 |50.41   |1.83     |57.08     |0.0  |1            |0      |0.0        |0.0          |0.02       |0.0        |0.0          |0.0        |0.0          |0.0           |0.0            |25.84        |
|2021-09-07 00:10:00|Room_3|20.0    |495.0   |27.95 |71.4    |6.7      |57.12     |89.82|1            |0      |0.0        |0.0          |0.05       |0.0        |908.44       |91.2       |1.61         |93.08         |0.0            |25.88        |
|2021-09-07 00:10:00|Room_4|20.0    |432.07  |26.18 |76.14   |3.73     |62.61     |18.18|2            |0      |0.0        |0.21         |0.0        |0.0        |45.14        |100.0      |7.74         |94.84         |0.0            |25.37        |
|2021-09-07 00:10:00|Room_5|59.13   |479.87  |28.07 |65.44   |6.67     |51.3      |240.33|2           |0      |0.0        |0.0          |0.02       |0.0        |1942.83      |72.98      |7.74         |94.84         |0.0            |25.37        |
|2021-09-07 00:15:00|Room_1|20.0    |422.4   |26.49 |74.74   |3.7      |49.69     |6.98 |0            |0      |0.01       |0.01         |0.04       |0.0        |0.0          |0.0        |0.0          |0.0           |0.0            |26.52        |
|2021-09-07 00:15:00|Room_2|20.0    |446.07  |23.25 |50.5    |1.7      |57.09     |0.0  |1            |0      |0.0        |0.0          |0.02       |0.0        |0.0          |0.0        |0.0          |0.0           |0.0            |25.84        |
|2021-09-07 00:15:00|Room_3|20.0    |484.0   |27.78 |69.09   |5.97     |56.98     |90.48|1            |0      |0.0        |0.0          |0.05       |0.0        |903.72       |91.2       |1.61         |93.08         |0.0            |25.88        |
|2021-09-07 00:15:00|Room_4|20.0    |426.07  |25.91 |74.77   |3.1      |62.78     |18.32|1            |0      |0.0        |0.0          |0.0        |0.0        |51.03        |100.0      |7.74         |94.84         |0.0            |25.37        |
|2021-09-07 00:15:00|Room_5|28.93   |471.97  |27.94 |64.72   |5.93     |50.88     |229.15|2           |0      |0.0        |0.0          |0.02       |0.0        |1942.83      |72.98      |7.74         |94.84         |0.0            |25.37        |
+-------------------+------+--------+--------+------+--------+---------+----------+-----+-------------+-------+-----------+-------------+-----------+-----------+-------------+-----------+-------------+-----------+-----------+-----------+
only showing top 20 rows
```

- After Spark DataFrame columns



```
--- 1-Hour Moving Average Table (Preview 20 rows) ---
+-------------------+-------------------+-------+--------+--------+--------+-------+---------+----------+-------+-------------+--------+-----------+-----------+-----------+-----------+-----------+-----------+-------------+---------------+-------------+
|start_time         |end_time           |room   |voc_ppb |co2_ppm |temp_c  |hum_pct|pm25_ugm3|noise_dba |lux    |wifi_dev_cnt |occ_cnt |fan_e_kwh  |light_e_kwh|plug_e_kwh |chw_e_kwh  |airflow_cmh|damper_pct |valve_pos_pct|supply_hum_pct |filter_press_pa|supply_temp_c|
+-------------------+-------------------+-------+--------+--------+--------+-------+---------+----------+-------+-------------+--------+-----------+-----------+-----------+-----------+-----------+-----------+-------------+---------------+-------------+-------------+
|2021-09-07 00:00:00|2021-09-07 01:00:00|Room_4 |20.02   |418.71  |25.71   |74.14  |2.78     |62.8      |35.18  |1.33         |0.0     |0.0        |0.05       |0.05       |0.0        |47.1       |100.0      |7.74         |94.84          |0.0          |25.41        |
|2021-09-07 00:00:00|2021-09-07 01:00:00|Room_1 |20.0    |428.07  |26.37   |73.13  |3.44     |50.43     |7.27   |0.0          |0.0     |0.01       |0.01       |0.04       |0.0        |0.0        |0.0        |0.0          |0.0            |0.0          |26.64        |
|2021-09-07 00:00:00|2021-09-07 01:00:00|Room_5 |59.34   |458.92  |27.73   |64.58  |5.45     |51.87     |238.54 |2.25         |0.0     |0.0        |0.0        |0.02       |0.0        |1942.83    |72.98      |7.74         |94.84          |0.0          |25.41        |
|2021-09-07 00:00:00|2021-09-07 01:00:00|Room_3 |20.0    |467.79  |27.52   |66.34  |4.98     |57.72     |56.33  |1.67         |0.0     |0.0        |0.0        |0.05       |0.0        |593.45     |93.14      |1.61         |93.08          |0.0          |25.77        |
|2021-09-07 00:00:00|2021-09-07 01:00:00|Room_2 |20.27   |439.92  |23.3    |49.99  |1.43     |57.24     |0.0    |1.0          |0.0     |0.0        |0.0        |0.02       |0.0        |0.0        |0.0        |0.0          |0.0            |0.0          |26.0         |
|2021-09-07 01:00:00|2021-09-07 02:00:00|Room_4 |63.62   |429.22  |25.22   |71.1   |1.57     |63.61     |190.57 |1.33         |0.0     |0.0        |0.05       |0.05       |0.0        |45.63      |100.0      |7.74         |94.84          |0.0          |25.53        |
|2021-09-07 01:00:00|2021-09-07 02:00:00|Room_5 |294.98  |461.93  |27.01   |64.2   |3.68     |55.36     |483.67 |2.75         |0.0     |0.0        |0.0        |0.02       |0.0        |1942.83    |72.98      |7.74         |94.84          |0.0          |25.53        |
|2021-09-07 01:00:00|2021-09-07 02:00:00|Room_2 |20.36   |428.4   |22.75   |48.7   |1.01     |57.44     |0.0    |1.0          |0.0     |0.0        |0.0        |0.02       |0.0        |0.0        |0.0        |0.0          |0.0            |0.0          |26.36        |
|2021-09-07 01:00:00|2021-09-07 02:00:00|Room_3 |20.0    |462.9   |27.04   |59.49  |2.63     |59.22     |109.51 |1.67         |0.0     |0.0        |0.0        |0.05       |0.0        |1.57       |100.0      |1.61         |93.08          |0.0          |25.7         |
|2021-09-07 01:00:00|2021-09-07 02:00:00|Room_1 |46.61   |578.2   |26.39   |67.69  |3.53     |57.32     |17.11  |0.0          |0.0     |0.0        |0.01       |0.04       |0.0        |0.0        |0.0        |0.0          |0.0            |0.0          |26.99        |
|2021-09-07 02:00:00|2021-09-07 03:00:00|Room_3 |20.0    |470.44  |26.9    |58.9   |3.12     |58.06     |120.44 |1.17         |0.0     |0.0        |0.0        |0.05       |0.0        |0.0        |100.0      |1.61         |93.08          |0.0          |26.01        |
|2021-09-07 02:00:00|2021-09-07 03:00:00|Room_1 |45.35   |542.08  |26.27   |68.0   |3.3      |55.53     |26.64  |0.0          |0.0     |0.0        |0.01       |0.04       |0.0        |0.0        |0.0        |0.0          |0.0            |0.0          |27.18        |
|2021-09-07 02:00:00|2021-09-07 03:00:00|Room_5 |147.82  |450.65  |26.94   |63.04  |2.64     |54.93     |606.97 |2.17         |0.0     |0.0        |0.02       |0.02       |0.0        |1942.83    |72.98      |7.74         |94.84          |0.0          |25.7         |
|2021-09-07 02:00:00|2021-09-07 03:00:00|Room_2 |20.47   |426.36  |22.35   |49.13  |0.97     |57.32     |3.06   |1.0          |0.0     |0.0        |0.0        |0.02       |0.0        |0.0        |0.0        |0.0          |0.0            |0.0          |26.62        |
|2021-09-07 02:00:00|2021-09-07 03:00:00|Room_4 |20.61   |434.96  |25.22   |70.74  |1.41     |63.6      |218.51 |1.5          |0.0     |0.0        |0.03       |0.05       |0.0        |44.41      |100.0      |7.74         |94.84          |0.0          |25.7         |
|2021-09-07 03:00:00|2021-09-07 04:00:00|Room_3 |20.0    |482.72  |26.9    |58.87  |3.92     |60.24     |116.77 |1.25         |0.0     |0.0        |0.0        |0.05       |0.0        |6.48       |100.0      |1.61         |93.08          |0.0          |26.14        |
|2021-09-07 03:00:00|2021-09-07 04:00:00|Room_5 |107.87  |446.41  |27.05   |62.35  |3.24     |55.51     |588.66 |2.25         |0.0     |0.0        |0.0        |0.02       |0.0        |1942.83    |72.98      |7.74         |94.84          |0.0          |25.82        |
|2021-09-07 03:00:00|2021-09-07 04:00:00|Room_1 |24.48   |455.29  |26.23   |74.49  |4.52     |52.77     |21.69  |0.0          |0.0     |0.0        |0.01       |0.04       |0.0        |0.0        |0.0        |0.0          |0.0            |0.0          |27.2         |
|2021-09-07 03:00:00|2021-09-07 04:00:00|Room_2 |20.58   |485.41  |22.65   |49.3   |1.37     |59.96     |82.04  |1.0          |0.0     |0.0        |0.0        |0.02       |0.0        |0.0        |0.0        |0.0          |0.0            |0.0          |26.71        |
|2021-09-07 03:00:00|2021-09-07 04:00:00|Room_4 |66.04   |460.16  |25.67   |69.91  |2.52     |63.54     |203.49 |1.33         |0.0     |0.0        |0.0        |0.05       |0.0        |35.49      |100.0      |7.74         |94.84          |0.0          |25.82        |
+-------------------+-------------------+-------+--------+--------+--------+-------+---------+----------+-------+-------------+--------+-----------+-----------+-----------+-----------+-----------+-----------+-------------+---------------+-------------+-------------+
only showing top 20 rows
```

5.3. Output Destinations

- /user/hduser/occupancy/batch_output/
    - /user/hduser/occupancy/batch_output/cleaned_overall_sensor
    - /user/hduser/occupancy/batch_output/moving_average_1hr
    - /user/hduser/occupancy/batch_output/{room}_sensor (e.g., Room_1_sensor, Room_2_sensor)
    - /user/hduser/occupancy/batch_output/agg_hourly_trends
    - /user/hduser/occupancy/batch_output/agg_daily_trends
    - /user/hduser/occupancy/batch_output/space_optimization
    - /user/hduser/occupancy/batch_output/environment_control
    - /user/hduser/occupancy/batch_output/environment_comfort_index
    - /user/hduser/occupancy/batch_output/avg_monthly_energy

6. Serving Layer: Document Database

6.1. MongoDB Data Model Design

Collection: room 1


```
{
  "_id": {
    "$oid": "69cfc0c72199961c99966baa"
  },
  "room": "Room_3",
  "area": "Quiet Study Area",
  "capacity": 15,
  "avg_occ": 1.67,
  "max_occ": 13,
  "average_usage_percent": 11.13,
  "category": "Low"
}
```


Collection: space_optimization


```
{
  "_id": {
    "$oid": "69cfc0c72199961c99966ba8"
  },
  "room": "Room_1",
  "area": "Reading Hall",
  "capacity": 40,
  "avg_occ": 2.19,
  "max_occ": 38,
  "average_usage_percent": 5.48,
  "category": "Low"
}
```


Collection: agg_hourly_trends


```
{
  "_id": {
    "$oid": "69cfc0a52199961c99953403"
  },
  "hour_of_day": 0,
  "avg_occupancy": 0.12,
  "category": "Low"
}
```


Collection: moving_average_1hr


```
{
  "_id": {
```

```
    "$oid": "69cfc0c12199961c99965ab0"
   },
   "start_time": {
    "$date": "2021-09-07T00:00:00.000Z"
   },
   "end_time": {
    "$date": "2021-09-07T01:00:00.000Z"
   },
   "room": "Room_4",
   "voc_ppb": 20.02,
   "co2_ppm": 418.71,
   "temp_c": 25.71,
   "hum_pct": 74.14,
   "pm25_ugm3": 2.78,
   "noise_dba": 62.8,
   "lux": 35.18,
   "wifi_dev_cnt": 1.33,
   "occ_cnt": 0,
   "fan_e_kwh": 0,
   "light_e_kwh": 0.05,
   "plug_e_kwh": 0.05,
   "chw_e_kwh": 0,
   "airflow_cmh": 47.1,
   "damper_pct": 100,
   "valve_pos_pct": 7.74,
   "supply_hum_pct": 94.84,
   "filter_press_pa": 0,
   "supply_temp_c": 25.41
  }
```


6.2. Supported Queries

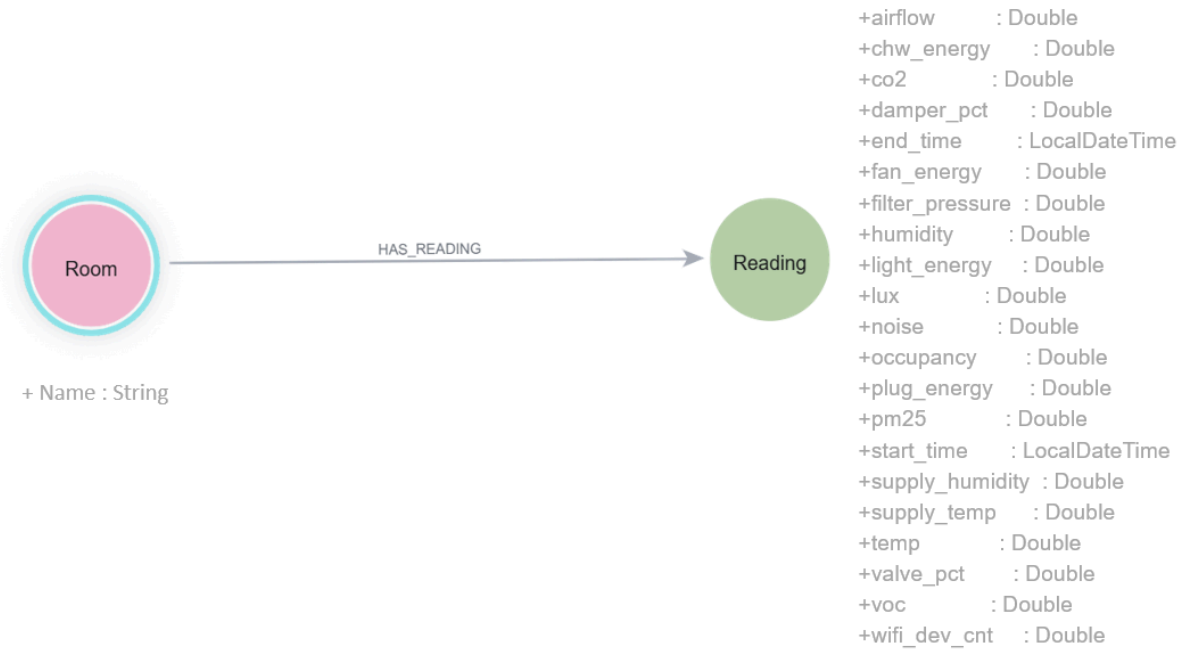
```
Busiest Day of the Week:
Day: Tuesday
Average Occupancy: 3.13
Category: Medium

Space Utilization Analysis (Rooms 1-5):

Most Used Areas :
Room: Room_3
Area: Quiet Study Area
Capacity: 15
Avg Occ: 1.67
Max Occ: 13
Usage%: 11.13
Category: Low
```

7. Serving Layer: Graph Database

7.1. Neo4j Graph Model Design



7.2. Supported Queries

```
===== Most Occupied Room =====
<Record room='Room_4' total_people=997.2899999999995>

===== Quietest Room =====
<Record room='Room_1' avg_noise=49.23318352059921>
<Record room='Room_5' avg_noise=51.989138576779055>
<Record room='Room_2' avg_noise=56.18644194756552>
```